

AP Statistics MCQ Practice Exam 06

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AP Statistics MCQ Practice Exam 6

PART A, QUESTIONS

APSTAT-MCQ6-001

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

At a streaming service, 0.35 of subscribers enable download notifications, and 0.12 enable both download notifications and dark mode. What is the probability that a randomly selected subscriber enables download notifications but does not enable dark mode?

- A) 0.12
- B) 0.18
- C) 0.23
- D) 0.47

APSTAT-MCQ6-002

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A high school wants to estimate the proportion of all students who are satisfied with the school lunch program. Administrators survey a simple random sample drawn only from students who purchased a school lunch that day. What is the primary concern?

- A) Students who brought lunch or did not eat are excluded from the sampling frame, creating undercoverage.
- B) The use of a simple random sample guarantees that the estimate is unbiased for all students.
- C) The sample is a voluntary-response sample because students purchased lunch.
- D) The study is an experiment because lunch choice was observed.

APSTAT-MCQ6-003

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

For the same difference in two population proportions, a 90% confidence interval is (0.010, 0.070), while a 99% confidence interval is (−0.007, 0.087). Which conclusion is justified?

- A) Both intervals provide evidence of a positive difference at their corresponding significance levels.
- B) The data provide evidence of a positive difference at the 0.10 level but not at the 0.01 level for a two-sided test.
- C) The population difference has a 90% probability of being between 0.01 and 0.07.
- D) The wider interval proves that the null hypothesis is true at the 0.01 level.

APSTAT-MCQ6-004

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

In a random sample of 25 devices, the standard error of the sample mean operating time is 1.2 hours. What is the sample standard deviation of operating times?

- A) 0.048
- B) 0.24
- C) 4.8
- D) 6.0

APSTAT-MCQ6-005

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Formulate Questions

Prompt:

A researcher wants to study the relationship between weekly practice time and performance score for a sample of musicians. What should the researcher do first to examine the form, direction, strength, and possible unusual features of the relationship?

- A) Construct separate pie charts for practice time and performance score.
- B) Compare the two sample means with a two-sample t-test.
- C) Make a scatterplot of performance score versus weekly practice time.
- D) Fit a regression line without first displaying the paired data.

APSTAT-MCQ6-006

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

The following ordered data are displayed in a stem-and-leaf plot, where 3|4 represents 34.

2 | 1 4

3 | 0 3 3 5

4 | 2

Which pair gives the median and range, respectively?

- A) Median 32; range 18
- B) Median 33; range 21
- C) Median 33; range 42
- D) Median 34; range 21

APSTAT-MCQ6-007

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A confidence interval for a population proportion is reported as (0.441, 0.519). The sample proportion is 0.480, and its estimated standard error is 0.020. What confidence level was used, approximately?

- A) 95%
- B) 90%
- C) 98%
- D) 99%

Questions APSTAT-MCQ6-008 to APSTAT-MCQ6-010 refer to the following stimulus.

A safety panel contains four alert buttons labeled 1, 2, 3, and 4. On each of two independent trials, one label is selected at random, with all four labels equally likely. Let X be the larger of the two selected labels.

APSTAT-MCQ6-008

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Two alert labels are selected independently and uniformly from 1, 2, 3, and 4, and X is the larger label. What is $P(X = 2)$?

- A) $1/16$
- B) $1/8$
- C) $1/4$
- D) $3/16$

APSTAT-MCQ6-009

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Two alert labels are selected independently and uniformly from 1, 2, 3, and 4, and X is the larger label. What is $P(X \geq 3)$?

- A) $7/16$
- B) $3/4$
- C) $13/16$
- D) $15/16$

APSTAT-MCQ6-010

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

Two alert labels are selected independently and uniformly from 1, 2, 3, and 4, and X is the larger label. Which value is a median of the probability distribution of X ?

- A) 1
- B) 2
- C) 3
- D) 4

APSTAT-MCQ6-011

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

Each participant uses keyboard layout A for one timed task and layout B for a comparable timed task, with the order randomized. If $d = \text{time with A} - \text{time with B}$ for each participant, which parameter should be the focus of inference?

- A) μ_d , the population mean of the within-participant time differences
- B) $\mu_A - \mu_B$ for two independent populations of participants
- C) $p_A - p_B$, the difference in population completion proportions
- D) σ_d , the population standard deviation of the within-participant time differences

APSTAT-MCQ6-012

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

The mean commute time for 50 employees is 18.4 minutes. Which statement must be true?

- A) Exactly half of the employees commute less than 18.4 minutes.
- B) At least one employee has a commute time of exactly 18.4 minutes.
- C) Most employees commute between 17.4 and 19.4 minutes.
- D) The sum of the 50 commute times is 920 minutes.

APSTAT-MCQ6-013

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

A least-squares regression line predicts a sound rating y from a machine setting x . Every sound rating is then recalibrated by adding 4 units, so $y^* = y + 4$. Which statement describes the regression of y^* on x ?

- A) The slope and correlation are unchanged, and the intercept increases by 4.
- B) The slope increases by 4, and the intercept and correlation are unchanged.
- C) The slope and intercept both increase by 4, and the correlation decreases.
- D) The slope is unchanged, the intercept increases by 4, and the correlation becomes $4r$.

APSTAT-MCQ6-014

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

A university wants to estimate the proportion of students who have experienced food insecurity, a sensitive topic. Which data-collection method is most likely to reduce response bias while preserving the ability to take a probability sample?

- A) Ask resident assistants to identify students they believe have experienced food insecurity.
- B) Interview students face-to-face and require them to state their names.
- C) Select students randomly and allow them to complete a confidential, self-administered questionnaire.
- D) Post an open survey link on social media and use everyone who responds.

APSTAT-MCQ6-015

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For randomly selected shipments, $P(\text{express}) = 0.30$ and $P(\text{damaged} \mid \text{express}) = 0.40$. What is $P(\text{express and damaged})$?

- A) 0.10
- B) 0.12
- C) 0.35
- D) 0.70

APSTAT-MCQ6-016

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A news organization tells interviewers to obtain 20 adults from each of four age groups at a downtown shopping district. Within each age group, interviewers approach whichever people seem easiest to contact. Which assessment is most appropriate?

- A) The design is a randomized block experiment because age groups are used.

- B) The equal quotas make the sample a simple random sample of city adults.
- C) The design prevents selection bias because every age group has the same sample size.
- D) The age quotas do not remove convenience selection within groups, so the sample can be biased.

APSTAT-MCQ6-017

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A one-sample t-test is performed on temperatures measured in degrees Celsius. Every observation and the null mean are converted to degrees Fahrenheit using $F = 1.8C + 32$. What happens to the t statistic?

- A) It is multiplied by 1.8.
- B) It increases by 32.
- C) It remains unchanged.
- D) It is divided by 1.8.

APSTAT-MCQ6-018

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A planning study suggests that a population proportion is about 0.20. What minimum sample size is needed so that a 95% confidence interval for the proportion has margin of error at most 0.06? Use $z^* = 1.96$ and the planning value 0.20.

- A) 171
- B) 170
- C) 683
- D) 1,068

APSTAT-MCQ6-019

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

A regression model predicts a building's daily energy use from outdoor temperature. The residual standard deviation is $s = 18$ kilowatt-hours. Which interpretation is appropriate?

- A) Outdoor temperature typically differs from its predicted value by about 18 degrees.
- B) Observed daily energy use typically differs from the line's predicted energy use by about 18 kilowatt-hours.
- C) Every building-day has a residual of exactly 18 kilowatt-hours.
- D) Outdoor temperature explains exactly 18% of the variation in daily energy use.

APSTAT-MCQ6-020

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

An instructor adds 5 points to every score on an exam. What happens to each student's percentile rank within the class?

- A) It increases by 5 percentile points.
- B) It is multiplied by 5.
- C) It changes only for students originally below the mean.
- D) It remains unchanged.

APSTAT-MCQ6-021

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

Rain is forecast independently on each of the next 14 days with probability 0.20 per day. Which random variable is appropriate for finding the probability that it rains on at least 5 of the days?

- A) X = the number of rainy days among the 14 days
- B) X = the day on which the fifth rainy day occurs
- C) X = the total rainfall amount over the 14 days
- D) X = the number of days until the first rainy day

APSTAT-MCQ6-022

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A nonprofit will survey a random sample of eligible voters after an outreach campaign. It wants to estimate the proportion who recall receiving a campaign message. Which parameter should be defined?

- A) x , the number of sampled voters who recall a message
- B) $\hat{p}$, the sample proportion of surveyed voters who recall a message
- C) p , the proportion of all eligible voters who recall receiving a campaign message
- D) μ , the mean number of messages received by sampled voters

APSTAT-MCQ6-023

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

A laboratory computes a 95% one-sided upper confidence bound of 4.8 milligrams per liter for the population mean nitrate concentration. The safety benchmark is $\mu < 5$ milligrams per liter. Which interpretation is best?

- A) At least 95% of individual nitrate measurements are below 4.8 milligrams per liter.
- B) There is a 95% probability that μ is below 4.8 milligrams per liter.
- C) The sample mean must equal 4.8 milligrams per liter because the bound is below 5.
- D) The procedure provides evidence at the corresponding one-sided 5% level that μ is below 5, but it does not describe every individual measurement.

APSTAT-MCQ6-024

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A dataset of $n > 1$ observations has mean $\bar{x}$ and positive sample standard deviation s . One additional observation equal to $\bar{x}$ is added. How do the sample mean and sample standard deviation change?

- A) The mean increases and the standard deviation decreases.
- B) The mean stays the same and the standard deviation decreases.
- C) The mean stays the same and the standard deviation stays exactly the same.
- D) The mean decreases and the standard deviation increases.

APSTAT-MCQ6-025

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Two independent random samples are used for a two-sample t procedure comparing population means. The sample standard deviations differ substantially. Which statement is correct about the usual unpooled two-sample t procedure?

- A) It is invalid unless the two sample standard deviations are exactly equal.
- B) It requires combining the two sample standard deviations into one pooled estimate.
- C) It does not require the two population variances to be equal, though unusual shapes and small samples can still be concerns.
- D) It becomes a paired t procedure whenever the sample sizes are equal.

APSTAT-MCQ6-026

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Let X be the number of 12 independently generated passwords that contain a special symbol, where each password has probability 0.30 of containing one. Calculator output gives $P(X \leq 2) = 0.253$ and $P(X \leq 5) = 0.961$. What is $P(3 \leq X \leq 5)$?

- A) 0.708
- B) 0.455
- C) 0.214
- D) 0.039

APSTAT-MCQ6-027

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

In a random sample of 100 items, 32 meet a standard. For testing $H_0: p = 0.40$ against $H_a: p < 0.40$, which is the approximate test statistic and p-value?

- A) $z = -0.80$ and p-value = 0.212
- B) $z = -1.33$ and p-value = 0.092

C) $z = -1.63$ and $p\text{-value} = 0.103$

D) $z = -1.63$ and $p\text{-value} = 0.051$

APSTAT-MCQ6-028

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A study of adolescent sleep asks parents to report the usual bedtime of their teenagers. What potential problem is most directly introduced by this choice?

- A) Placebo effect, because parents know they are reporting sleep behavior
- B) Measurement bias from proxy reports that may not accurately reflect the teenagers' actual bedtimes
- C) Confounding caused by randomly assigning parents to report bedtimes
- D) Undercoverage of parents who know their teenagers' bedtimes

APSTAT-MCQ6-029

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Collect Data

Prompt:

A park manager wants to compare mean trail-use counts on summer weekdays with mean trail-use counts on winter weekdays. Which design best supports inference about the difference?

- A) Randomly select weekdays from each season and record the trail-use count on every selected day.
- B) Use the busiest summer weekday and the quietest winter weekday.
- C) Record summer counts at the main entrance and winter counts at a side entrance.
- D) Let visitors choose which season's count they want to contribute to.

APSTAT-MCQ6-030

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

Two independent studies test the same null value for a population proportion. Both have the same observed difference $\hat{p} - p_0 = 0.04$, but the first uses $n = 200$ and the second uses $n = 2,000$. The

second study has a much smaller p-value. Which interpretation is best?

- A) The effect observed in the second study is ten times as large.
- B) The second study proves that the null value is impossible.
- C) The estimated effect is the same, but the larger sample gives a smaller standard error and stronger statistical evidence.
- D) The smaller p-value shows that the second study used a less representative sample.

APSTAT-MCQ6-031

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

For each patient, a researcher records the difference between heart rate after and before a breathing exercise. Which display is most appropriate for examining the distribution of the patient-level differences?

- A) A segmented bar chart of before and after status
- B) A dotplot of the numerical differences
- C) A scatterplot with no patient pairing indicated
- D) A two-way table of heart-rate categories

Questions APSTAT-MCQ6-032 to APSTAT-MCQ6-034 refer to the following stimulus.

For 20 city blocks, x is tree-canopy coverage (percent) and y is afternoon sidewalk temperature (degrees Celsius). The least-squares line is $\hat{y} = 38 - 0.20x$. The sample means are $\bar{x} = 35$ and $\bar{y} = 31$. A twenty-first block is added with $x = 35$ and $y = 31$.

APSTAT-MCQ6-032

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For the original 20 blocks, $\hat{y} = 38 - 0.20x$, $\bar{x} = 35$, and $\bar{y} = 31$. What happens to the least-squares regression line after the block (35, 31) is added?

- A) Its slope becomes more negative because the new point is below all points with $x = 35$.
- B) Its slope becomes 0 because the new point is at the sample means.
- C) Its intercept increases to 38.2 because the sample size increases.
- D) The fitted line remains $\hat{y} = 38 - 0.20x$.

APSTAT-MCQ6-033

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For the original 20 blocks, $\hat{y} = 38 - 0.20x$, $\bar{x} = 35$, and $\bar{y} = 31$. What happens to the sum of squared residuals after the block (35, 31) is added?

- A) It must increase because there is one more observation.
- B) It decreases by 31^2 .
- C) It remains unchanged.
- D) It becomes 0.

APSTAT-MCQ6-034

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

For the original 20 blocks, $\hat{y} = 38 - 0.20x$, $\bar{x} = 35$, and $\bar{y} = 31$. How should the added point (35, 31) be characterized relative to the fitted relationship?

- A) It has residual 0 and is not a high-leverage point because its x-value is at the center of the x-values.
- B) It has residual 7 and is a high-leverage point because its y-value is large.
- C) It has residual -7 and must substantially change the slope.
- D) It has residual 0 but is a high-leverage point because every point on the line has high leverage.

APSTAT-MCQ6-035

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Collect Data

Prompt:

A meteorologist wants to examine whether a high-ozone day makes a high-ozone day more likely on the following day. Which plan best supplies data for comparing the relevant conditional probabilities while limiting overlap among observations?

- A) Select only high-ozone days and count how many are followed by high-ozone days.

- B) Randomly select many well-separated dates, record ozone status on each selected date and the next date, and compare next-day high-ozone rates for the two initial-day statuses.
- C) Record the total number of high-ozone days in one year without preserving their order.
- D) Ask residents whether they remember two high-ozone days occurring in succession.

APSTAT-MCQ6-036

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A one-proportion z-test of $H_0: p = 0.25$ with $n = 400$ produces $z = 2.31$. What sample proportion was observed, approximately?

- A) 0.20
- B) 0.25
- C) 0.28
- D) 0.30

APSTAT-MCQ6-037

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

For a random sample of 9 package masses, the sum of the squared deviations from the sample mean is 162 square kilograms. What is the sample variance?

- A) 20.25 square kilograms
- B) 9.0 square kilograms
- C) 4.5 square kilograms
- D) 2.25 square kilograms

APSTAT-MCQ6-038

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A matched-pairs study enrolls 30 participants and records a measurement before and after a program. Four participants have no after-program measurement and are excluded. What degrees of freedom should be used for the paired t procedure based on the complete pairs?

- A) 29
- B) 26
- C) 25
- D) 24

APSTAT-MCQ6-039

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

Two fair six-sided dice are rolled, and X is the absolute difference between the two results. What does $P(X = 0) = 1/6$ represent?

- A) At least one die shows 0 on one sixth of rolls.
- B) The two dice show the same number on 6 of the 36 equally likely ordered outcomes.
- C) The sum of the two dice is 0 on one sixth of rolls.
- D) The second die is determined by the first die on every roll.

APSTAT-MCQ6-040

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A small company attempts a census of all employees about workplace injuries, but some employees do not respond and others misreport minor injuries. Which statement is correct?

- A) A census eliminates sampling variability but does not eliminate nonresponse or measurement error.
- B) A census automatically eliminates all forms of bias.
- C) Nonresponse can matter only when a random sample, rather than a census, is used.
- D) Misreporting is sampling error and disappears when every employee is contacted.

APSTAT-MCQ6-041

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A test uses $H_0: p = 0.50$ and $H_a: p > 0.50$. The observed sample proportion is 0.45, giving a negative z statistic. Which conclusion is appropriate?

- A) Reject H_0 because the sample proportion differs from 0.50.
- B) Reject H_0 because every negative z statistic is significant.
- C) Conclude that $p < 0.50$ because that is the stated alternative.
- D) Fail to reject H_0 ; the result is in the direction opposite to H_a and the right-tailed p -value is large.

APSTAT-MCQ6-042

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

For a fixed numerical dataset, consider $S(c) = \sum(x_i - c)^2$. At which value of c is $S(c)$ minimized?

- A) The minimum observation
- B) The median
- C) The mean
- D) The maximum observation

PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ6-001, Correct: C, Correct Answer: 0.23, Explanation: Subtract the overlap from the probability of notifications: $P(\text{notifications and not dark mode}) = 0.35 - 0.12 = 0.23$.

APSTAT-MCQ6-002, Correct: A, Correct Answer: Students who brought lunch or did not eat are excluded from the sampling frame, creating undercoverage., Explanation: Random selection within a restricted frame represents only lunch purchasers. It does not give all students a chance to be selected.

APSTAT-MCQ6-003, Correct: B, Correct Answer: The data provide evidence of a positive difference at the 0.10 level but not at the 0.01 level for a two-sided test., Explanation: The 90% interval excludes the null value 0, whereas the 99% interval includes 0. This corresponds to rejecting at $\alpha = 0.10$ but not at $\alpha = 0.01$ for two-sided tests.

APSTAT-MCQ6-004, Correct: D, Correct Answer: 6.0, Explanation: For a sample mean, $SE = s/\sqrt{n}$. Thus $s = 1.2\sqrt{25} = 1.2(5) = 6.0$ hours.

APSTAT-MCQ6-005, Correct: C, Correct Answer: Make a scatterplot of performance score versus weekly practice time., Explanation: A scatterplot is the appropriate first display for two quantitative variables measured on the same individuals. It reveals form, direction, strength, and unusual

observations before a model is fitted.

APSTAT-MCQ6-006, Correct: B, Correct Answer: Median 33; range 21, Explanation: The seven values are 21, 24, 30, 33, 33, 35, and 42. The fourth value is 33, and the range is $42 - 21 = 21$.

APSTAT-MCQ6-007, Correct: A, Correct Answer: 95%, Explanation: The margin of error is $0.519 - 0.480 = 0.039$, so $z^* = 0.039/0.020 = 1.95$, approximately 1.96. That critical value corresponds to about 95% confidence.

APSTAT-MCQ6-008, Correct: D, Correct Answer: 3/16, Explanation: $X = 2$ occurs when both labels are at most 2 but not both are at most 1. Therefore $P(X = 2) = (2/4)^2 - (1/4)^2 = 4/16 - 1/16 = 3/16$.

APSTAT-MCQ6-009, Correct: B, Correct Answer: 3/4, Explanation: The complement of $X \geq 3$ is that both selected labels are at most 2. Thus $P(X \geq 3) = 1 - (2/4)^2 = 1 - 1/4 = 3/4$.

APSTAT-MCQ6-010, Correct: C, Correct Answer: 3, Explanation: $P(X \leq 2) = (2/4)^2 = 0.25$, while $P(X \leq 3) = (3/4)^2 = 0.5625$. The cumulative probability first reaches at least 0.5 at $X = 3$.

APSTAT-MCQ6-011, Correct: A, Correct Answer: μ_d , the population mean of the within-participant time differences, Explanation: Both measurements come from the same participant, so the observations are paired. Inference should be based on the population mean of the defined within-participant differences.

APSTAT-MCQ6-012, Correct: D, Correct Answer: The sum of the 50 commute times is 920 minutes., Explanation: Since $\bar{x} = \Sigma x/n$, the sum is $n \bar{x} = 50(18.4) = 920$ minutes. A mean does not determine the median, any individual value, or a narrow concentration of values.

APSTAT-MCQ6-013, Correct: A, Correct Answer: The slope and correlation are unchanged, and the intercept increases by 4., Explanation: Adding 4 to every response shifts all points and predictions vertically by 4. It does not change standardized association or the amount y changes per unit of x .

APSTAT-MCQ6-014, Correct: C, Correct Answer: Select students randomly and allow them to complete a confidential, self-administered questionnaire., Explanation: Random selection supports representation, and confidential self-administration reduces the social pressure and interviewer effects that can distort answers to sensitive questions.

APSTAT-MCQ6-015, Correct: B, Correct Answer: 0.12, Explanation: By the multiplication rule, $P(\text{express and damaged}) = P(\text{express})P(\text{damaged} \mid \text{express}) = 0.30(0.40) = 0.12$.

APSTAT-MCQ6-016, Correct: D, Correct Answer: The age quotas do not remove convenience selection within groups, so the sample can be biased., Explanation: Quotas control the number sampled in each age category, but interviewers still choose convenient individuals. Without random selection within groups, inclusion probabilities are unknown and selection bias can remain.

APSTAT-MCQ6-017, Correct: C, Correct Answer: It remains unchanged., Explanation: Both $\bar{x} - \mu_0$ and s are multiplied by 1.8 under the conversion, while n is unchanged. The common factor cancels in $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$.

APSTAT-MCQ6-018, Correct: A, Correct Answer: 171, Explanation: The requirement is $n \geq (1.96^2)(0.20)(0.80)/(0.06^2) \approx 170.74$. Sample size must be rounded up, giving 171.

APSTAT-MCQ6-019, Correct: B, Correct Answer: Observed daily energy use typically differs from the line's predicted energy use by about 18 kilowatt-hours., Explanation: The residual standard deviation describes the typical vertical size of prediction errors in the response variable's units. It is not a slope, a percentage of explained variation, or an exact residual for every case.

APSTAT-MCQ6-020, Correct: D, Correct Answer: It remains unchanged., Explanation: Adding the same constant to all scores preserves their order and all relative ranks, so percentile ranks do not change.

APSTAT-MCQ6-021, Correct: A, Correct Answer: X = the number of rainy days among the 14 days, Explanation: The event "at least 5 of the days" concerns a count of successes in a fixed number of independent trials, so X should count rainy days among the 14.

APSTAT-MCQ6-022, Correct: C, Correct Answer: p , the proportion of all eligible voters who recall receiving a campaign message, Explanation: The parameter is a numerical characteristic of the entire target population. The sample count and sample proportion are statistics used to estimate it.

APSTAT-MCQ6-023, Correct: D, Correct Answer: The procedure provides evidence at the corresponding one-sided 5% level that μ is below 5, but it does not describe every individual measurement., Explanation: The upper bound 4.8 lies below the benchmark 5, so the corresponding one-sided test rejects $\mu = 5$ in favor of $\mu < 5$ at $\alpha = 0.05$. The bound concerns the population mean, not a stated percentage of individual values.

APSTAT-MCQ6-024, Correct: B, Correct Answer: The mean stays the same and the standard deviation decreases., Explanation: Adding a value equal to the old mean leaves the mean and the sum of squared deviations unchanged. The sample-variance denominator increases from $n - 1$ to n , so the positive standard deviation decreases.

APSTAT-MCQ6-025, Correct: C, Correct Answer: It does not require the two population variances to be equal, though unusual shapes and small samples can still be concerns., Explanation: The unpooled procedure estimates each sample's contribution to the standard error separately. Equal variances and equal sample sizes are not requirements.

APSTAT-MCQ6-026, Correct: A, Correct Answer: 0.708, Explanation: The event $3 \leq X \leq 5$ is obtained by subtracting cumulative probabilities: $P(X \leq 5) - P(X \leq 2) = 0.961 - 0.253 = 0.708$.

APSTAT-MCQ6-027, Correct: D, Correct Answer: $z = -1.63$ and $p\text{-value} = 0.051$, Explanation: $z = (0.32 - 0.40)/\sqrt{[0.40(0.60)/100]} \approx -1.63$. For the left-tailed alternative, the standard normal area below -1.63 is about 0.051.

APSTAT-MCQ6-028, Correct: B, Correct Answer: Measurement bias from proxy reports that may not accurately reflect the teenagers' actual bedtimes, Explanation: Parents may not observe the true bedtime or may report it inaccurately. This is a measurement problem caused by using proxies, not random assignment or undercoverage as described.

APSTAT-MCQ6-029, Correct: A, Correct Answer: Randomly select weekdays from each season and record the trail-use count on every selected day., Explanation: Random selection within each season

gives representative independent samples for estimating and comparing the two seasonal population means.

APSTAT-MCQ6-030, Correct: C, Correct Answer: The estimated effect is the same, but the larger sample gives a smaller standard error and stronger statistical evidence., Explanation: The two point estimates are equally far from the null value. Increasing n decreases the standard error, increasing the magnitude of the test statistic and decreasing the p -value without changing the estimated effect size.

APSTAT-MCQ6-031, Correct: B, Correct Answer: A dotplot of the numerical differences, Explanation: The paired differences form one quantitative variable. A dotplot displays its shape, center, spread, and unusual values directly.

APSTAT-MCQ6-032, Correct: D, Correct Answer: The fitted line remains $\hat{y} = 38 - 0.20x$., Explanation: The added point is exactly $(\bar{x}, \bar{y})$, which lies on the existing line because $38 - 0.20(35) = 31$. It contributes zero to both centered cross-products and centered squared x deviations, so the fitted line is unchanged.

APSTAT-MCQ6-033, Correct: C, Correct Answer: It remains unchanged., Explanation: The existing fitted values and residuals do not change, and the added point lies exactly on the line, so its residual and squared residual are both zero.

APSTAT-MCQ6-034, Correct: A, Correct Answer: It has residual 0 and is not a high-leverage point because its x -value is at the center of the x -values., Explanation: Its predicted value is $38 - 0.20(35) = 31$, equal to its observed value. Leverage depends on unusual x -position, and $x = \bar{x}$ is central rather than extreme.

APSTAT-MCQ6-035, Correct: B, Correct Answer: Randomly select many well-separated dates, record ozone status on each selected date and the next date, and compare next-day high-ozone rates for the two initial-day statuses., Explanation: Recording both statuses for randomly selected, well-separated day pairs permits comparison of $P(\text{next high} \mid \text{initial high})$ with $P(\text{next high} \mid \text{initial not high})$ while reducing overlap among pairs.

APSTAT-MCQ6-036, Correct: D, Correct Answer: 0.30, Explanation: Rearranging $z = (\hat{p} - 0.25)/\sqrt{[0.25(0.75)/400]}$ gives $\hat{p} = 0.25 + 2.31\sqrt{(0.1875/400)} \approx 0.25 + 0.05 = 0.30$.

APSTAT-MCQ6-037, Correct: A, Correct Answer: 20.25 square kilograms, Explanation: Sample variance divides the sum of squared deviations by $n - 1$. Thus $s^2 = 162/(9 - 1) = 162/8 = 20.25$ square kilograms.

APSTAT-MCQ6-038, Correct: C, Correct Answer: 25, Explanation: Only the 26 participants with both measurements contribute paired differences. A paired t procedure on 26 differences has $df = 26 - 1 = 25$.

APSTAT-MCQ6-039, Correct: B, Correct Answer: The two dice show the same number on 6 of the 36 equally likely ordered outcomes., Explanation: An absolute difference of 0 occurs exactly for $(1,1)$, $(2,2)$, ..., $(6,6)$. Those 6 matching outcomes among 36 equally likely ordered pairs give probability $6/36 = 1/6$.

APSTAT-MCQ6-040, Correct: A, Correct Answer: A census eliminates sampling variability but does not eliminate nonresponse or measurement error., Explanation: Attempting to contact the full population

removes random sampling variability, but missing responses and inaccurate measurements can still bias the result.

APSTAT-MCQ6-041, Correct: D, Correct Answer: Fail to reject H_0 ; the result is in the direction opposite to H_a and the right-tailed p-value is large., Explanation: For H_a : $p > 0.50$, evidence lies in the positive-z tail. A negative statistic places most of the normal curve to its right, so the p-value is large.

APSTAT-MCQ6-042, Correct: C, Correct Answer: The mean, Explanation: The arithmetic mean is the least-squares center: the sum of squared deviations from c is minimized when $c = \bar{x}$.

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ6

- APSTAT-MCQ6-001 to 005: Unit 2 streaming-service notification and dark-mode settings subtract their intersection from a marginal probability; Unit 1 a school-lunch satisfaction SRS drawn only from purchasers exposes frame undercoverage; Unit 3 nested 90% and 99% intervals for a difference in proportions lead to different two-sided decisions; Unit 4 device operating times recover s from the standard error of $\bar{x}$ and n ; Unit 5 paired musician practice time and performance call for a scatterplot before fitting. Misconceptions include subtracting the wrong event, treating random selection inside an incomplete frame as representative, assigning posterior probability to interval endpoints, omitting $\sqrt{n}$, and fitting a model before displaying the data.
- APSTAT-MCQ6-006 to 010: Unit 1 a seven-value stem-and-leaf plot gives its median and range; Unit 3 interval endpoints, $\hat{p} = 0.480$, and $SE = 0.020$ recover $z^* \approx 1.96$ and a 95% confidence level; Unit 2 a three-item stimulus defines X as the maximum of two independent uniform alert labels and finds $P(X = 2)$ by cumulative differencing, $P(X \geq 3)$ by complement, and the distribution median. Misconceptions include misreading stems, confusing full width with margin of error, counting only one ordering, complementing the wrong event, and treating a one-trial label median as the median of X .
- APSTAT-MCQ6-011 to 015: Unit 4 randomized within-person keyboard-layout times define μ_d ; Unit 1 a 50-employee mean commute converts to the total time; Unit 5 a four-unit shift of every sound response changes only the regression intercept; Unit 3 a probability sample with confidential self-administration reduces sensitive-question response bias; Unit 2 express and damaged shipments multiply a marginal probability by a conditional probability. Misconceptions include treating paired measurements as independent groups, assuming a mean must occur, changing slope or correlation under a location shift, replacing random sampling with an open link, and reversing a conditional.
- APSTAT-MCQ6-016 to 020: Unit 1 downtown age-quota interviewing still has convenience selection within groups; Unit 4 Celsius-to-Fahrenheit conversion of both observations and null leaves a t statistic invariant; Unit 3 a planning value $p^* = 0.20$ with 95% margin 0.06 determines $n = 171$; Unit 5 a building-energy regression interprets residual standard deviation as typical prediction-error size in kilowatt-hours; Unit 1 a five-point shift of all exam scores preserves percentile ranks. Misconceptions include treating quotas as random selection, shifting a standardized statistic, rounding n down, confusing residual spread with slope or r^2 , and adding a constant to ranks.
- APSTAT-MCQ6-021 to 025: Unit 2 next-14-day rain defines a fixed-trial count variable rather than a waiting time; Unit 3 campaign-message recall distinguishes population p from x and $\hat{p}$; Unit 4 a 95% one-sided upper bound of 4.8 for mean nitrate concentration supports $\mu < 5$ at the corresponding 5% level; Unit 1 adding an observation at $\bar{x}$ keeps the mean and reduces the positive sample standard deviation; Unit 4 the unpooled independent two-sample t procedure does not require equal variances. Misconceptions include choosing a waiting-time variable, labeling a statistic as a parameter, applying a mean bound to individual observations or assigning posterior probability, overlooking the changed sample-variance denominator, and imposing a pooled-variance assumption.

- APSTAT-MCQ6-026 to 030: Unit 2 a binomial count of 12 independently generated passwords uses two supplied cumulative probabilities to find $P(3 \leq X \leq 5)$; Unit 3 a left-tailed one-proportion test from $n = 100$ and $\hat{p} = 0.32$ calculates z and a one-tail p -value; Unit 1 parents reporting teenagers' bedtimes create proxy measurement bias; Unit 4 independent random weekday samples compare mean trail use between seasons; Unit 3 holds $\hat{p} - p_0$ fixed while a tenfold increase in n reduces standard error and p -value. Misconceptions include retaining the lower cumulative probability rather than subtracting it, doubling a one-sided p -value, calling proxy error undercoverage, selecting extreme days or changing entrances, and treating a smaller p -value as a larger estimated effect.
- APSTAT-MCQ6-031 to 035: Unit 1 a dotplot displays quantitative within-patient heart-rate differences; Unit 5 a three-item tree-canopy and sidewalk-temperature stimulus adds the centroid point $(35, 31)$ on $\hat{y} = 38 - 0.20x$, leaving the fitted line and SSE unchanged and giving zero residual with low leverage; Unit 2 ozone dependence data use randomly selected well-separated initial/next-day pairs to compare conditional rates. Misconceptions include discarding pair structure, assuming every added point changes a line or SSE, confusing a central on-line point with an influential point, and conditioning only on high-ozone initial days.
- APSTAT-MCQ6-036 to 040: Unit 3 solves a one-proportion z formula backward for $\hat{p}$; Unit 1 nine package masses use $\sum(x_i - \bar{x})^2/(n - 1)$ to calculate sample variance; Unit 4 a 30-person before/after study with four incomplete pairs uses 26 differences and $df = 25$; Unit 2 the absolute difference of two fair dice interprets $P(X = 0)$ through the six matching ordered outcomes; Unit 1 an employee injury census removes sampling variability but retains nonresponse and measurement error. Misconceptions include using $\hat{p}$ instead of p_0 in the test standard error, dividing a sample sum of squares by n , counting incomplete cases or individual measurements as pairs, confusing difference zero with a zero die result or zero sum, and assuming a census is bias-free.
- APSTAT-MCQ6-041 to 042: Unit 3 a negative z statistic supplies no right-tailed evidence for $H_a: p > 0.50$; Unit 1 the arithmetic mean minimizes $\sum(x_i - c)^2$ for a fixed numerical dataset. Misconceptions include rejecting merely because $\hat{p}$ differs from p_0 or reversing the stated alternative, and confusing the least-squares center with the median.
- Unit counts for APSTAT-MCQ6: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: maximum of two uniformly selected alert labels, APSTAT-MCQ6-008 to 010; adding a centroid point on a tree-canopy regression line, APSTAT-MCQ6-032 to 034.